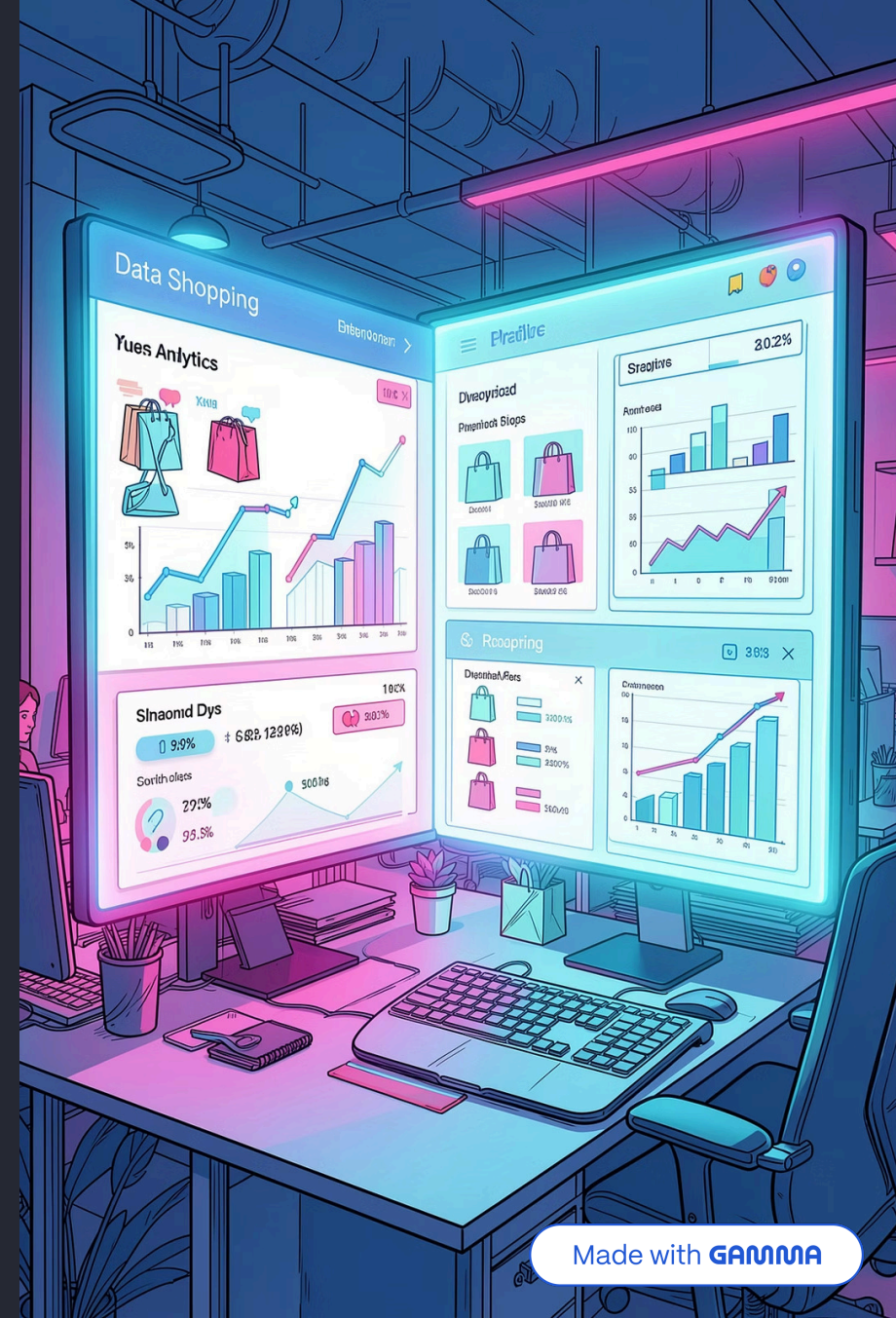


# Customer Behaviour Analysis

End-to-end business analysis of 3,900 retail transactions — combining Python, PostgreSQL, and Power BI to uncover purchasing trends, customer segments, and growth opportunities.

PREPARED BY: RUMAVA BISWAS



# Business Objectives

## Clean & Prepare

Handle missing values, remove duplicates, and standardize categorical data.

## Explore Demographics

Analyze age, gender, location, and purchase frequency distributions.

## SQL Business Queries

Answer key business questions using aggregation, window functions, and CASE statements.

## Power BI Dashboard

Build an interactive executive dashboard with KPIs, slicers, and visualizations.



# Dataset Overview

3,900

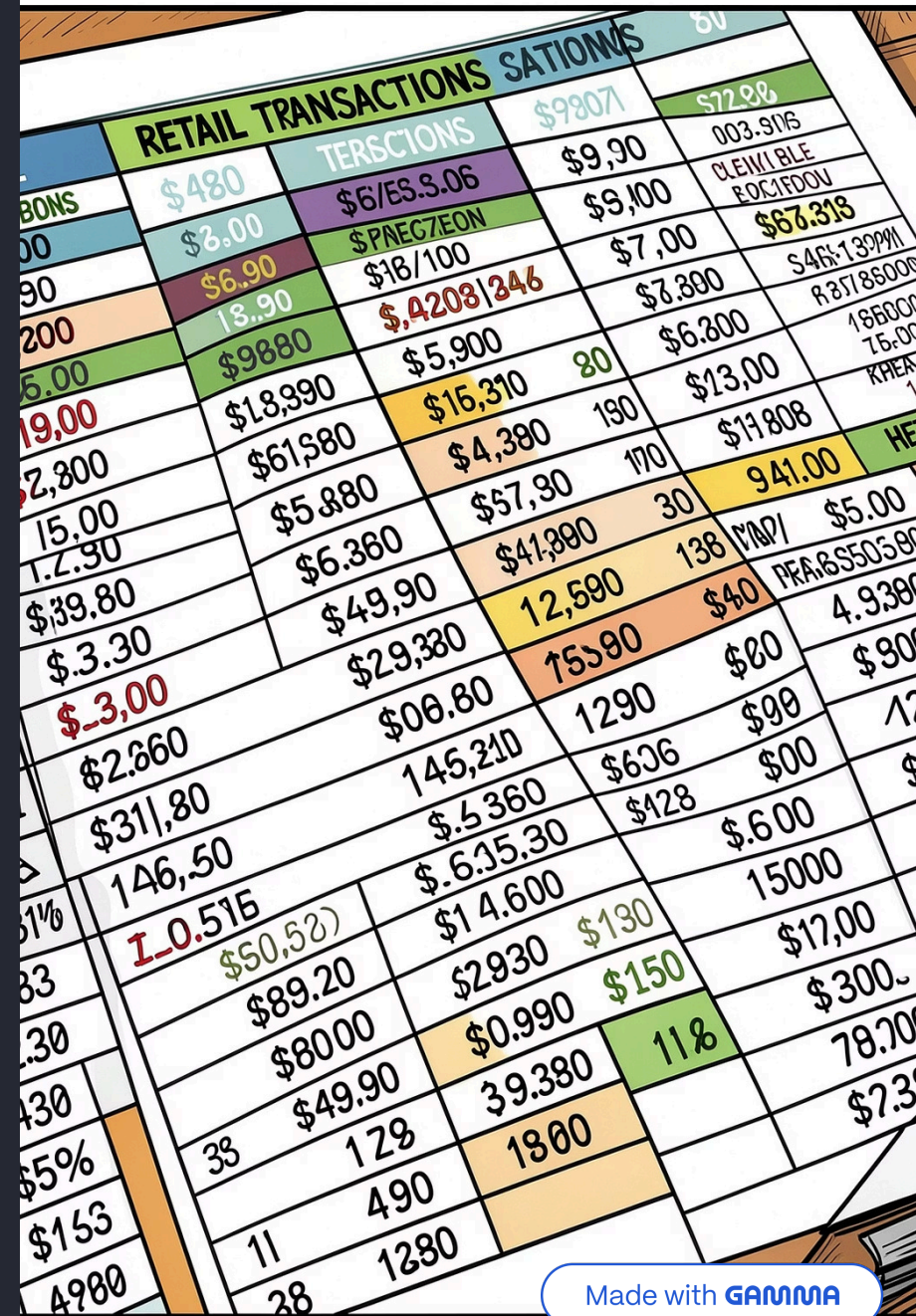
Transactions

18

Attributes

Key fields include Purchase Amount, Review Rating, Subscription Status, Shipping Type, Payment Method, and Frequency of Purchases. Customer ages range from **18 to 70**.

- Average Purchase Amount: **~\$60 USD**
- Average Review Rating: **~3.75**
- Average Previous Purchases: **~25**



| RETAIL TRANSACTIONS |          |        |           |
|---------------------|----------|--------|-----------|
| DATE                | AMOUNT   | RATING | STATUS    |
| 01/01/2020          | \$480    | 5.0    | Completed |
| 01/02/2020          | \$2.00   | 3.5    | Pending   |
| 01/03/2020          | \$6.90   | 4.2    | Completed |
| 01/04/2020          | \$18.90  | 4.8    | Completed |
| 01/05/2020          | \$9880   | 5.0    | Completed |
| 01/06/2020          | \$18,390 | 4.5    | Completed |
| 01/07/2020          | \$61,580 | 4.8    | Completed |
| 01/08/2020          | \$5,880  | 4.5    | Completed |
| 01/09/2020          | \$6,360  | 4.5    | Completed |
| 01/10/2020          | \$49,90  | 4.5    | Completed |
| 01/11/2020          | \$29,330 | 4.5    | Completed |
| 01/12/2020          | \$06.80  | 4.5    | Completed |
| 01/13/2020          | \$311,80 | 4.5    | Completed |
| 01/14/2020          | 146,50   | 4.5    | Completed |
| 01/15/2020          | 1,0515   | 4.5    | Completed |
| 01/16/2020          | \$89.20  | 4.5    | Completed |
| 01/17/2020          | \$8000   | 4.5    | Completed |
| 01/18/2020          | \$49.90  | 4.5    | Completed |
| 01/19/2020          | 129      | 4.5    | Completed |
| 01/20/2020          | 490      | 4.5    | Completed |
| 01/21/2020          | 1280     | 4.5    | Completed |

# Data Cleaning

## Missing Values

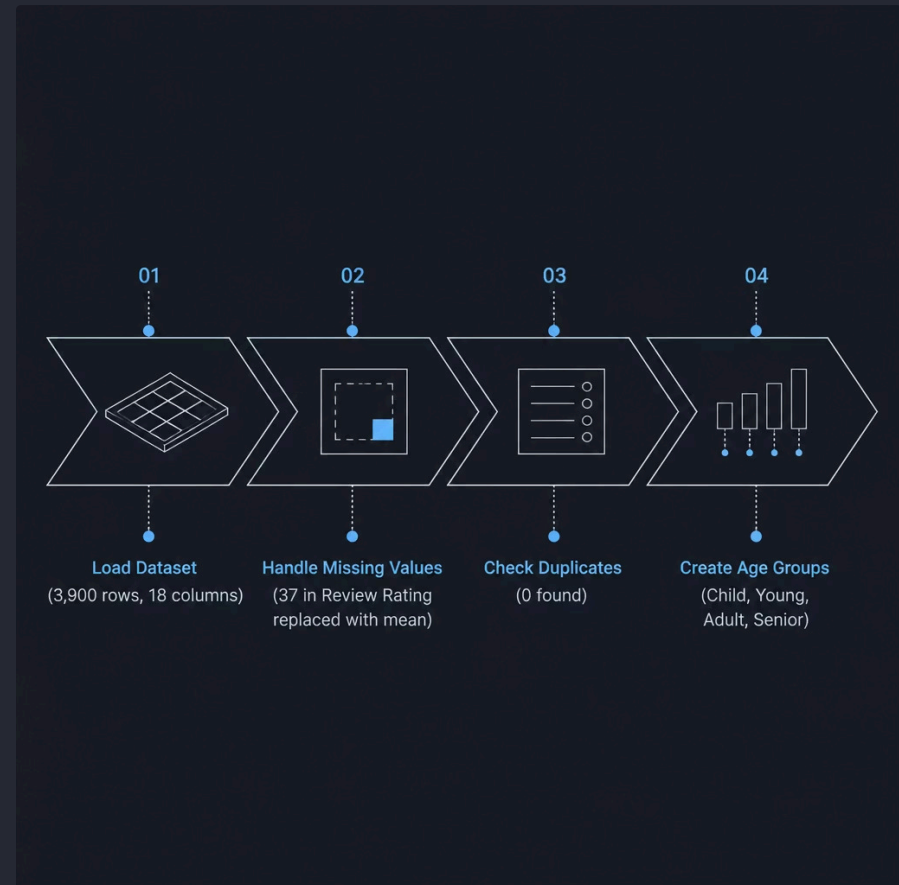
Review Rating had **37 missing values**, replaced using the column mean (3.75). All other columns were complete.

## Duplicates

Zero duplicate rows found. Categorical columns were explored for unique value distributions.

## Age Grouping

Ages segmented into: **Child, Young, Adult, Senior** using `pd.cut()`.



# Exploratory Data Analysis

## Gender Distribution

Male: **2,652** (68%) · Female: **1,248** (32%)

## Age Groups

Adult: **2,645** · Senior: **1,105** · Young: **150** · Child: **0**

## Purchase Frequency

Most even distribution across Weekly (539) to Every 3 Months (584).

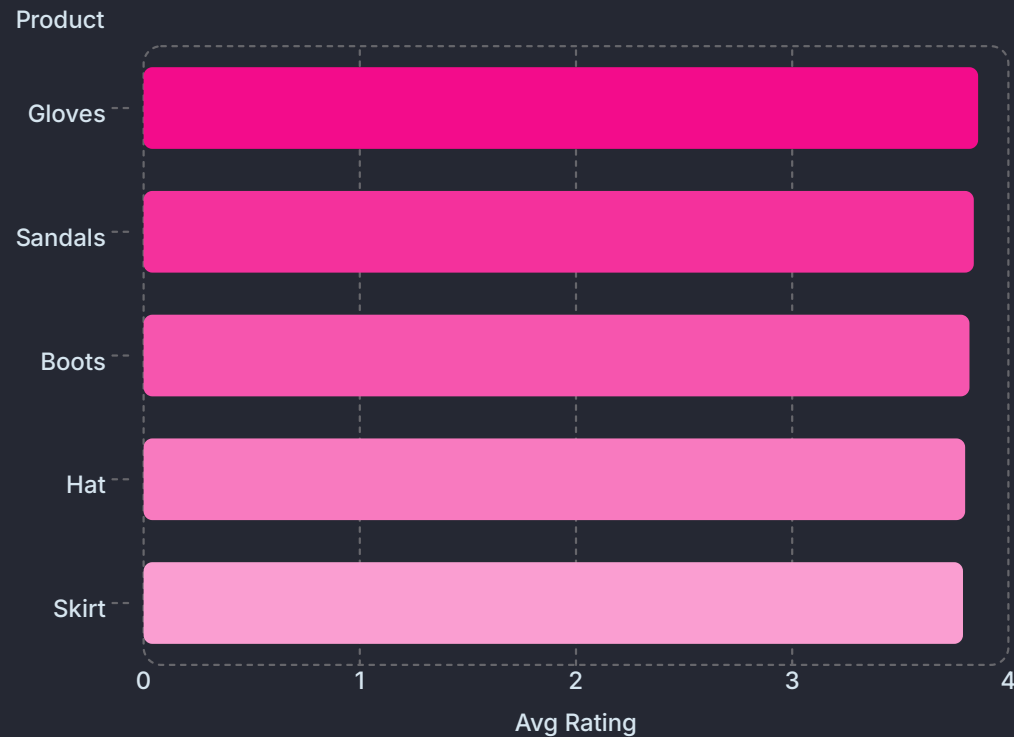
## Top Items

Blouse & Pants lead at **171** each; 18 product types represented.

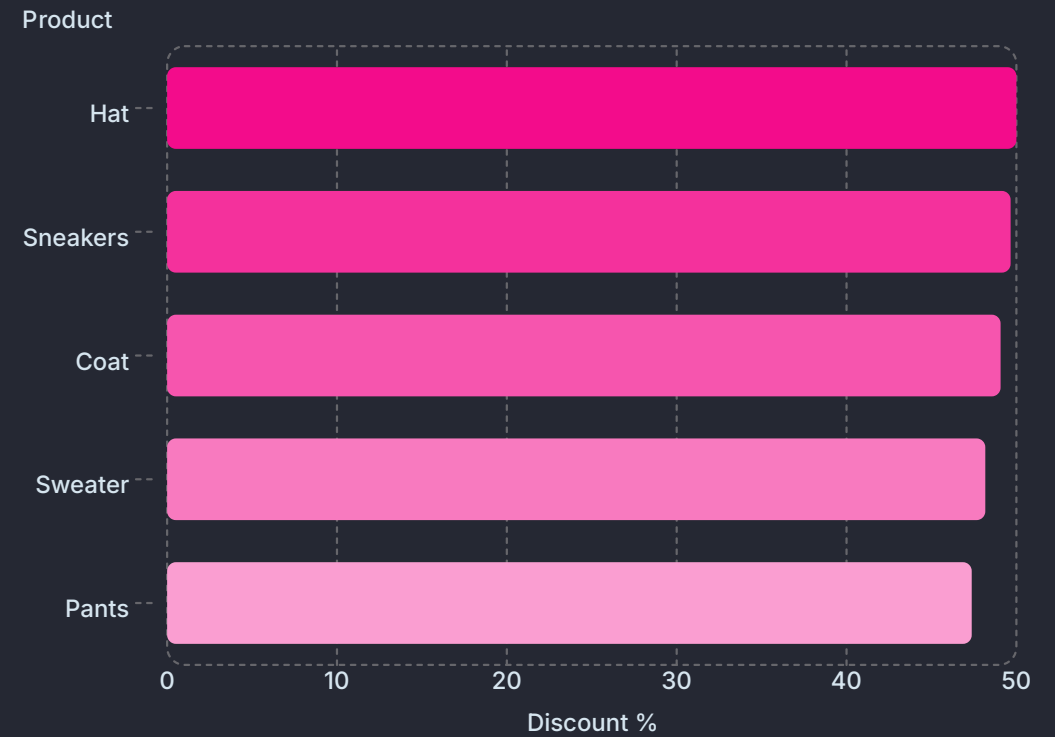


# SQL Business Insights — Ratings & Discounts

## Top 5 Products by Review Rating

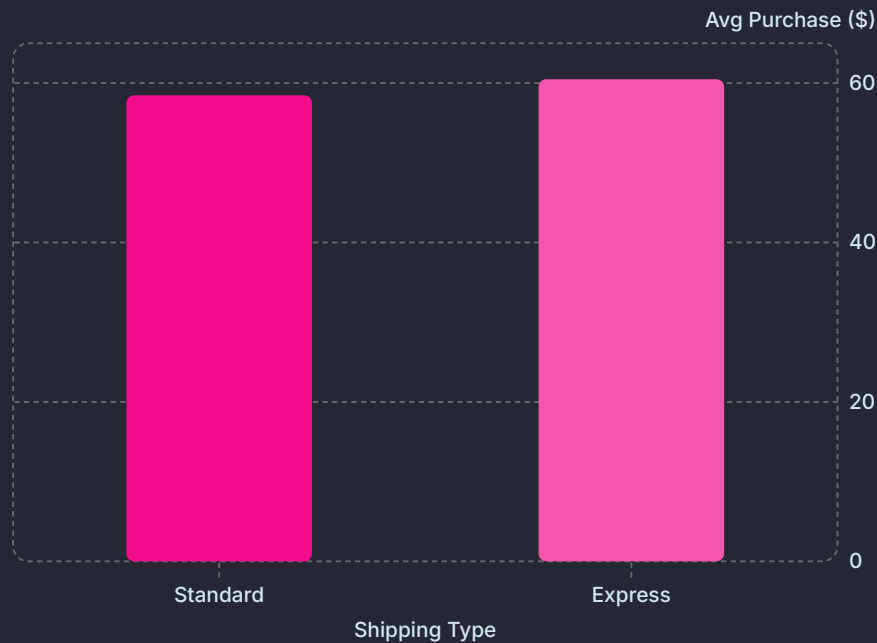


## Top 5 Products by Discount %

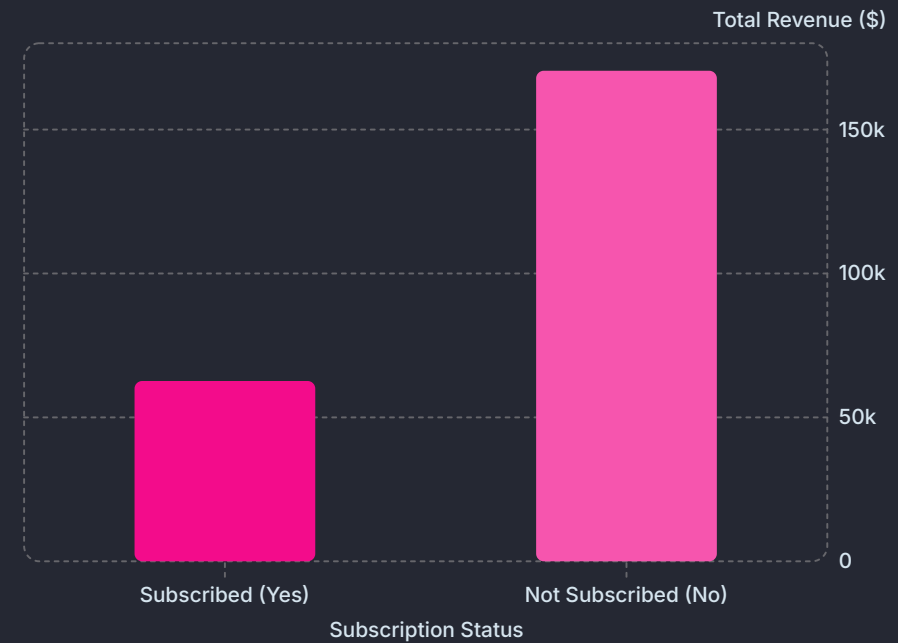


# SQL Business Insights — Shipping & Subscriptions

Average Spend by Shipping Type



Revenue: Subscribed vs Non-Subscribed



Non-subscribers generate **2.7× more revenue**. Express shipping customers spend **~\$2 more** on average than Standard.

# Advanced SQL Techniques

1

## Customer Segmentation

CASE statements classify customers as **New**, **Returning**, or **Loyal** based on purchase history.

2

## Category Ranking

DENSE\_RANK() window function identifies the **top 3 purchased products** within each category.

3

## Repeat Buyer Analysis

Filtered customers with **5+ previous purchases** and cross-referenced subscription status for retention insights.



Payment method analysis revealed **Venmo** as the most used and **Bank Transfer** as the least used.

# Power BI Dashboard

233K

Total Previous Purchases

4K

Total Customers

3.75

Avg Review Rating

\$59.76

Avg Purchase Amount

## Dashboard Features

- KPI cards with slicers for Category, Gender, Shipping, and Subscription
- Revenue & Sales by Category — **Clothing leads**
- Revenue by Age Group — Adults & Seniors dominate
- Subscription rate: **27% Yes / 73% No**
- Top locations: Montana, California, Idaho

# Key Insights & Recommendations

## Key Insights

- Clothing contributes the highest sales; Jewelry, Blouse & Pants are top items
- Adult customers dominate the base; avg rating ~3.75
- Express shipping customers spend slightly more
- Loyal customers form the largest segment

## Business Recommendations

- Increase inventory for top-selling categories
- Launch loyalty offers for returning customers
- Improve subscription benefits to boost the 27% rate
- Promote high-rated products like Gloves & Sandals
- Focus marketing on the adult customer segment
- Optimize shipping offers to improve conversion